Chapter 4: Lateral biogeography of thermophilic endospores in North Atlantic sediment

4.1 Abstract

Detection of dormant thermospores in permanently cold sediments can be used to study the potential vectors causing geographic dispersal of microorganisms in the marine environment. Surface sediments from nine locations in the North Atlantic were pasteurized (80°C for 1 hour) and incubated at 50°C to enrich thermospores in the sediment. Thermospore OTUs were detected in all nine sediments. Nine of the 41 thermospore OTUs detected were identified in more than one sediment and OTU 2 (affiliated with *Symbiobacterium ostreiconchae*) and OTU 46 (affiliated with *Bacillus thermoamylovorans*) appeared in incubations of the most sediments, which were three and five different sediments respectively. An analysis of the thermospores shared between different sediments demonstrated that while the North Atlantic subpolar gyre is a primary driver of North Atlantic circulation, this current is not exclusively responsible for the dispersal of the identified thermospores.

4.2 Introduction

Microbial biogeography is the study of microorganisms across geographic space and over geologic time. Hanson *et al*. (2012) identified selection, mutation, drift, and dispersal as the processes central to shaping microbial biogeography. The process of dispersal, defined by Hanson *et al*. (2012) as bacterial movement to and successful colonization, including activity and reproduction, at a new location, can be divided into two components, bacterial movement and successful colonization. Bacteria possessing the endospore phenotype have an increased ability to survive passive movement beyond their original location because the robust nature of the endospore phenotype allows the endospore to withstand stresses that would otherwise have destroyed the vegetative cell. Thermophilic bacteria, those with temperature optima between 45°C and 80°C, will only grow in environments with temperatures within their growth range and thus their growth is limited to these environments. Thermospores can be dispersed widely because they remain viable after exposure to environmental stress as endospores, yet their successful establishment in a new location is limited to warm environments. For these reasons, thermospores are a useful model organism for the study of the movement aspect of microbial dispersal specifically. Their movement is passive and unaffected by confounding factors of selection, mutation, and drift because the thermospores are dormant and resistant to environmental stressors during their dispersal. In this way, thermospores are similar to abiotic particles except they are biotic and are able to germinate and colonize a warm temperature environment should that environment be encountered during their passive dispersal.

When a warm environment is not encountered and thermospores are deposited in an environment unsuitable for germination, they contribute to the dormant microbial seed bank and overall biodiversity of that site (Lennon and Jones, 2011). The microbial seed bank is composed of the microorganisms in a particular location that are metabolically inactive and not reproducing. These organisms do not participate in local biogeochemical cycling and so do not alter the geochemical properties of the environment nor do they impact the growing populations in that environment. They may however become active when the conditions change and become suitable for their

activity and growth. For thermospores lying dormant in a cool location, an increase in temperature (e.g. in an experiment) may cause germination and growth so that the active community structure and dominant microbial activities shift to reflect that of the thermophilic endospore-forming bacteria.

The ability to withstand environmental stress and remain viable during passive dispersal implies that endospore-forming bacteria should be very good dispersers and could potentially be cosmopolitan (present everywhere) in the dormant microbial seed bank globally. Can endospore-forming bacteria, possibly the most equipped of all microorganisms for passive dispersal, show dispersal limitation across large geographic distances? Thermospores, with their tolerance to environmental stress and inability to germinate in cool environments characteristic to most of the earth's surface biosphere, offer a biological mechanism and tool for addressing this question.

It has long been observed that thermophilic microorganisms can be enriched from cool environments (McBee and McBee, 1955; Bartholomew and Paik, 1966; Isaksen *et al*., 1994; Marchant *et al*., 2002; Lee *et al*., 2005). Hubert *et al*. (2009) showed that thermophilic endospore-forming bacteria can be enriched from Arctic sediment and that the flux of these thermophiles is continuous to the Arctic environment. Microorganisms are dispersed thousands of kilometers through the atmosphere (Bovallius *et al*., 1980; Griffin *et al*., 2002). Likewise, ocean currents have been shown to disperse microorganisms (Wilkins *et al*., 2013) and thermospores have been detected and enumerated in the water column above sediments in which they are found (de Rezende *et al*., 2013; Volpi *et al*., 2017) providing a link between the dispersal vector of ocean currents, and the seabed sediment. Some microbial dispersal limitation in the marine environment has been shown. Agogué *et al*. (2011) demonstrated that bacterial communities cluster according to water masses and Salazar *et al*. (2016) showed that microorganisms attached to particles show oceanic basin specificity. Additionally, Müller *et al*. (2014) illustrated that thermospores found in sediments globally exhibit both cosmopolitanism and dispersal limitation, which is the uneven probability of dispersal (Hanson, *et al*., 2012).

The North Atlantic subpolar gyre is a dominant feature of North Atlantic ocean circulation. It is composed of a branch of the North Atlantic Current that veers west after

crossing the Mid-Atlantic Ridge at a latitude of 48°N. It then follows the northern edges of the seas and basins of the North Atlantic south of Iceland, Greenland, and Baffin Island towards the Labrador slope where the water cools and descends, forming the deep water of the North Atlantic (Marzocchi *et al*., 2015). This counterclockwise current is called the North Atlantic subpolar gyre. A branch of the North Atlantic Current veers northeast after crossing the Mid-Atlantic Ridge becoming the Norwegian Current that moves into the Nordic sea and meets the western edges of Svalbard (Nøst and Isachsen, 2003; Marzocchi *et al*., 2015).

The Mid-Atlantic Ridge systems have been considered a warm temperature source of thermospores (Hubert *et al*., 2009; de Rezende *et al*., 2013, Müller *et al*., 2014). These geochemically rich systems support organisms from all domains of life (Cerqueira *et al*., 2013) and thermophiles with optima near 50°C, including relatives to those identified in this study, may inhabit suitable hot environments in areas adjacent to hydrothermal vents. Deming and Baross (1993) have suggested that fluids emanating from hydrothermal plumes, which may rise hundreds of metres before becoming neutrally buoyant (Lupton *et al*., 1985), may carry thermophiles to the water column above. Other studies have suggested that these plumes may contribute to the dispersal of vent-associated organisms (Jackson *et al*., 2010; Dick *et al*., 2013).

The North Atlantic Current connects hydrothermal environments, such as the Lost City hydrothermal system found near the Mid-Atlantic Ridge, to other areas of the North Atlantic and may be a dispersal vector of thermospores in the water column. In this study, the dispersal of thermospores in North Atlantic sediment was assessed using 50°C microcosm incubations of surface sediment from nine different locations in the North Atlantic. To asses the likelihood of thermospore dispersal from the Mid-Atlantic Ridge to sediment at locations connected to the North Atlantic subpolar gyre, and to assess thermospore endemism and cosmopolitanism in the North Atlantic, thermospore OTUs occurring in 50°C incubations from nine different North Atlantic sediment sites were compared.

4.3 Materials and Methods

4.3.1 North Atlantic sediment locations

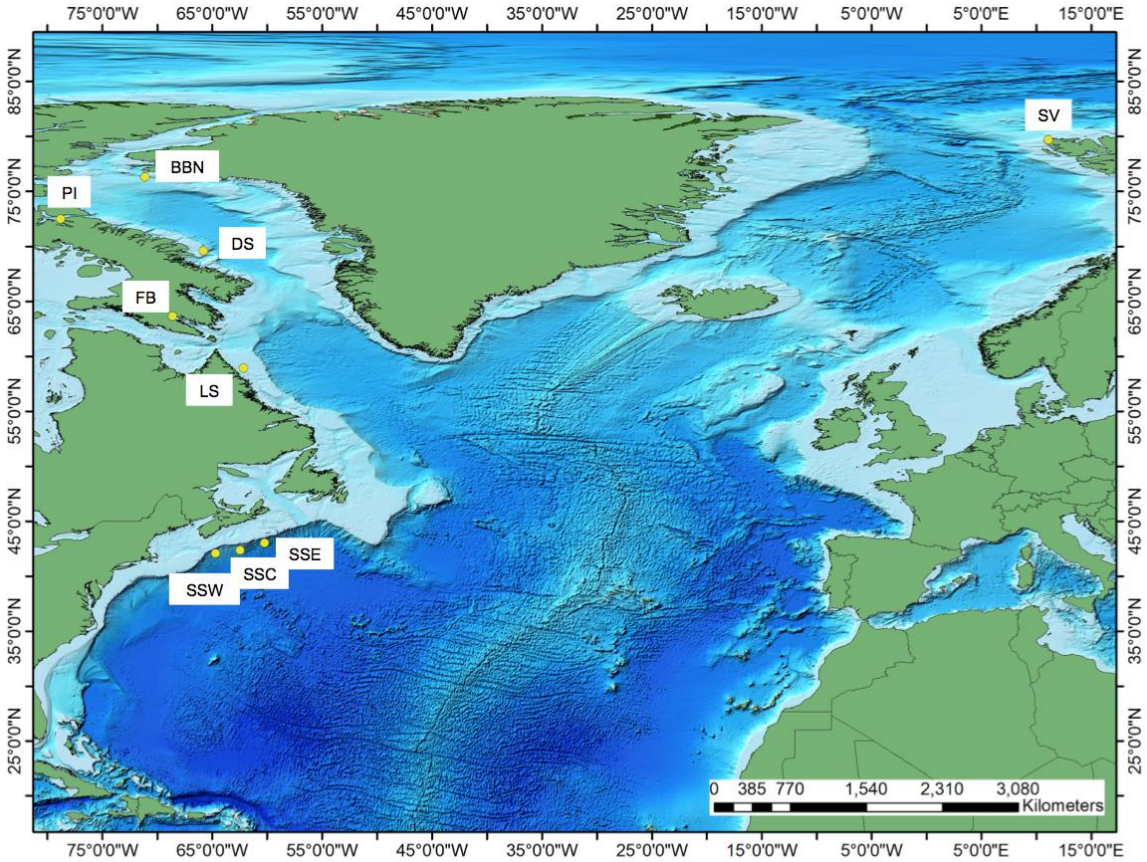

Figure 4.1: North Atlantic surface sediment used in 50°C thermospore incubations. Sites indicated in the map are Scotian Slope West (SSW), Scotian Slope Center (SSC), Scotian Slope East (SSE), Labrador Shelf (LS), Frobisher Bay (FB), Davis Strait (DS), Pond Inlet (PI), Baffin Bay North (BBN), and Svalbard (SV).

Thermospore biogeography studies use the germination of thermospores to circumvent the difficulties associated with detecting thermophilic endospore-forming bacteria based on molecular methods alone (de Rezende *et al*., 2017). Nine sites in the North Atlantic were used for incubation experiments in this study. Figure 4.1 shows the locations of all nine sediments and Table 2.1 lists the geographic details of each site. Each site was designated as one of two groups: sites that were connected (Connected

Sites, abbreviated to CS) to the North Atlantic subpolar gyre and sites that were not connected (Not Connected Sites, abbreviated to NCS) to the North Atlantic subpolar gyre (Table 4.3). Five of the nine sediment sites were considered connected (CS) to the North Atlantic subpolar gyre (Scotian Slope West, Scotian Slope Centre, Scotian Slope East, Labrador Shelf, and Svalbard). The other four of the nine sediment sites were considered not connected (NCS) to the North Atlantic subpolar gyre (Frobisher Bay, Davis Strait, Pond Inlet, and Baffin Bay North). Svalbard was designated as connected to the North Atlantic subpolar gyre (CS) despite not being geographically connected because it is connected to the Norwegian Current, a north extending arm of the North Atlantic Current that branches after the North Atlantic Current crosses the Mid-Atlantic Ridge, thus the Norwegian Current possesses a similar history to the North Atlantic subpolar gyre current.

The sediment from each site was stored anoxically at 4°C prior to 50°C incubation. Sediment from sites Baffin Bay North and Davis Strait were stored in glass containers with airtight lids. Sediments from the seven other sites were stored in anoxic plastic bags sealed with binder fasteners.

4.3.2 High temperature incubation

Anoxic 50°C sediment incubations were established as described in section 2.2. Microcosms were amended with the organic acids formate, lactate, acetate, succinate, propionate, and butyrate, to a final concentration of 1 mM each. An error in a stock solution led to microcosms of sediment from station Baffin Bay North and Davis Strait receiving organic acids at concentrations 10-fold below that of the seven other sediment incubations (0.1 mM) which was only detected after the conclusion of the incubation experiment. Svalbard sediment incubations analyzed in this study are the same incubations discussed in Chapter 3 as positive controls. These Svalbard microcosms contained 1mM of ethanol in addition to the 1 mM of the organic acids formate, lactate, acetate, succinate, propionate, and butyrate; the eight other sediment incubations in this study lacked ethanol in their medium. Differences in electron donor concentrations, as is the case with incubations of sediment from Baffin Bay North and Davis Strait, and

electron donor type, as is the case with incubations of sediment from Svalbard, mean that a particular OTU found in incubations of a certain electron donor mixture or concentration cannot be considered absent in incubations of the other condition even if it is not identified in that other condition. OTU identification is further discussed in section 2.4.3.

Pasteurization before incubation and subsampling over the incubation period occurred as per section 2.2. Subsamples were mixed 1:1 (v/v) with a 2% solution of zinc acetate before centrifugation and separation of the supernatant and pellet fractions before storage at -20°C.

4.3.3 Sulfate and organic acid measurement

Sulfate concentration of the subsamples taken during the course of the experiment was measured using both the turbidimetric method (Nemati *et al*., 2001) (Svalbard samples only) and ion chromatography (Dionex ICS-5000, Thermo Scientific) as described in section 2.3. Organic acid concentrations were measured by UHPLC as described in section 2.3 using the Acclaim™ Organic Acid LC Column (5µm, 4 × 250 mm; Thermo Scientific) for the Svalbard incubation samples and the Aminex HPX-87H column (5µm, 7.8 × 300 mm, Bio Rad) for all other samples. Organic acid measurements were taken at Day 0 and Day 26, 28 or 53 in all sediment incubations, but if sulfate reduction was observed at an intermediate time point, then organic acids were also measured at that time point. The microcosms were amended with the organic acids formate, lactate, succinate, propionate, butyrate, and acetate, but acetate concentrations were not analyzed because the addition of zinc acetate to subsamples during the subsampling procedure to squelch sulfate oxidization interfered with the detection of acetate in the subsample. The Acclaim™ Organic Acid LC Column (5µm, 4 × 250 mm; Thermo Scientific) used for measurement of organic acids in the Svalbard samples poorly measured formate and thus formate was unable to be measured with confidence in the Svalbard samples.

4.3.4 16S rRNA gene amplicon sequencing and analysis

DNA extraction of the Svalbard incubations from days 0 and 7 were done using the in-house method described by Foght *et al*. (2015). All other DNA extractions were done using the DNeasy Powersoil Kit (Qiagen). 16S rRNA gene amplification and sequencing was done as described in section 2.4.2 for all samples. Community analysis and thermospore OTU identification were done as described in section 2.4.3.

4.4 Results

4.4.1 Microbial activity and community structure during 50°C sediment incubation

Sulfate and organic acid concentrations were monitored over the 28-, 30-, or 53-day 50°C incubation to monitor microbial activity during that time. Sulfate reduction occurred upon 50°C incubation in many of the microcosms although often not in each biological replicate. A decrease in organic acid(s) over the high temperature incubation was also observed in most microcosms. Microcosms of sediment from Davis Strait and Baffin Bay North received only a tenth of the organic acid concentration as microcosms of sediment from all other sites. As such, organic acid concentrations remained very low in the Davis Strait and Baffin Bay North microcosms over the 28-day incubation.

All known thermospores belong to the bacterial phylum *Firmicutes*, thus an enrichment of thermospore-forming bacteria during the 50°C incubation is predicted to reveal a simultaneous increase in *Firmicutes* relative abundance. The relative abundance of *Firmicutes* was measured through a phylum-level analysis of 16S rRNA gene libraries as described in section 2.4.3. 16S rRNA gene libraries from before incubation (Day 0) and after 50°C incubation (Day 7, Day 14, and or Day 53) were made to assess the community-level response at the phylum-level and OTU-level to anoxic incubation at 50°C under the conditions described in section 2.2 and 2.3.2.

A summary of sulfate reduction, organic acid utilization, and increase in *Firmicutes* relative abundance is shown in Table 4.1. In each microcosm that displays microbial activity as evidenced by sulfate reduction and/or a decrease in organic acid concentration, an increase in *Firmicutes* relative abundance is observed in the microbial community. In some cases an increase in *Firmicutes* relative abundance is also observed in replicates where no sulfate reduction or decrease in organic acid concentration has occurred (Table 4.1 and Figures S4.4, S4.8, S4.9, S4.11) but there is no instance of a replicate undergoing sulfate reduction or a decrease in organic acid concentration without a corresponding increase in *Firmicutes* relative abundance (Table 4.1). These results suggest that sulfate reduction and decreases in organic acid concentration, which are measures of microbial activity, are associated to increases in *Firmicutes* relative abundance in these microcosms. Notably, the increase in *Firmicutes*

relative abundance occurs in at least one replicate from incubations of sediment from each of the nine sites in this study.

The results of incubations of each sediment site are discussed below. A summary of the results is shown in Table 4.1. A summary of the OTUs identified at each sediment site is shown in Table 4.2.

Table 4.1: Summary of the sulfate reduction, organic acid utilization, and increase in *Firmicutes* relative abundance after 50°C incubation. The symbols "+" and "-" represent a positive or negative result respectively in each replicate for the occurrence of sulfate reduction, organic acid utilization, or increase in *Firmicutes* relative abundance over the incubation period. The first symbol represents the result in the first replicate, the second represents the result in the second replicate, etc.

* Scotian Slope West sediment incubations had six biological replicates.

** Triplicate samples were pooled for 16S rRNA sequencing.

(Table is shown on the next page.)

Sediment site	Measure	Result
Scotian Slope West*	Sulfate reduction	− − − − − −
	Organic acid utilization	− − + + − +
	Increase in *Firmicutes* relative abundance	− − + + − +
Scotian Slope Centre	Sulfate reduction	− − +
	Organic acid utilization	− − +
	Increase in *Firmicutes* relative abundance	− − +
Scotian Slope East	Sulfate reduction	− − −
	Organic acid utilization	− − +
	Increase in *Firmicutes* relative abundance	+ − +
Svalbard	Sulfate reduction	+ + +
	Organic acid utilization	+ + +
	Increase in *Firmicutes* relative abundance	+ + +
Labrador Shelf	Sulfate reduction	− − +
	Organic acid utilization	+ + +
	Increase in *Firmicutes* relative abundance	+ + +
Baffin Bay North**	Sulfate reduction	− − −
	Organic acid utilization	− − −
	Increase in *Firmicutes* relative abundance	+
Pond Inlet	Sulfate reduction	− − −
	Organic acid utilization	− − −
	Increase in *Firmicutes* relative abundance	+ + −
Davis Strait**	Sulfate reduction	− − −
	Organic acid utilization	− − −
	Increase in *Firmicutes* relative abundance	+
Frobisher Bay	Sulfate reduction	− + +
	Organic acid utilization	+ + +
	Increase in *Firmicutes* relative abundance	+ + +

Table 4.2: Summary of the thermospore OTUs identified in each sediment after 50°C incubation. A "+" in a coloured box identifies a thermospore OTU as present in incubations of that sediment. A blue box represents a thermospore OTU that was identified in an incubation that also showed sulfate reduction and a decrease in organic acid concentration. A green box represents a thermospore OTU that was identified in an incubation that showed a reduction in organic acid concentration but no sulfate reduction. A grey box represents a thermospore OTU that was identified in an incubation that did not show either sulfate reduction or a decrease in organic acid concentration.

	Sediment site								
Thermospore OTU	SSW	SSC	SSE	LS	FB	DS	PI	BBN	Sv
2			+		+			+	
4					+				
5			+						
12	+	+							
14	+								
17					+				
18					+				
21			+						
34				+					+
44					+				
46			+	+		+	+	+	
55					+				
74					+				
89				+					+
121									+
136								+	+
145									+
168				+					
196									+
212				+					
216					+				
219									+
220				+					
224	+								
244									+
339								+	

Table continued on next page.

	Sediment site								
Thermospore OTU	SSW	SSC	SSE	LS	FB	DS	PI	BBN	Sv
359									+
411									+
412									+
469				+					+
514	+								
522					+				
532									+
562									+
610								+	
642					+			+	
748								+	
1146					+				
1236								+	
1535									+
2250					+				
4497			+						

4.4.1.1 Scotian Slope West

Sulfate reduction was not observed in incubations of Scotian Slope West (SSW) sediment, but decreases in organic acid concentration and increases in *Firmicutes* relative abundance were observed in 3 of 6 replicates (Table 4.1). OTUs 12, 14, 224, and 514 were identified after incubation at 50°C (Table 4.2). Figure S4.1 shows the sulfate concentration, organic acid concentration, phylum-level community structure and relative abundances of *Firmicutes* OTUs in sediment incubated from the Scotian Slope West site. Sulfate reduction did not occur over 11 days of incubation at 50°C, but decreases in formate and succinate concentrations in replicates 3, 4, and 6, and a decrease in butyrate concentration in replicate 4, suggest microbial activity in these replicates at 50°C. Replicates 3, 4, and 6 also show an increase in *Firmicutes* relative abundance after 50°C incubation (Figure S4.1c) while the phylum-level structure of replicates 1, 2 and 5 does not change suggesting that the changes in sulfate and organic acid concentration in replicates 3, 4 and 6 may be driven by *Firmicutes*

enrichment. After 50°C incubation replicates 3 and 6 are dominated by a taxon OTU 12 (*Thermicanus aegyptius*) while replicate 4 is dominated by OTU 14 (*Vulcanibacillus modesticaldus*) (Figure S4.2d; Table S4.1).

4.4.1.2 Scotian Slope Centre

OTU 12 was the only thermospore identified in Scotian Slope Centre sediment incubations (Table 4.2). Only replicate 3 from incubations (50°C) of this sediment shows sulfate reduction after 28 days (Figure S4.3b). This replicate also shows decreases in formate, lactate, succinate, propionate, and butyrate (Figure S4.3c) and its phylum-level community structure shifts to being dominated by *Firmicutes* after 50°C incubation (Figure S4.3d). After incubation at 50°C the 16S rRNA gene amplicon library of replicate 3 is dominated OTU 12 (*Thermicanus aegyptius*) (Table S4.1) which is also seen in incubations of Scotian Slope West sediment (Figure S4.1d). Conversely, replicates 1 and 2 do not display any changes and thus show no evidence of thermospore enrichment.

4.4.1.3 Scotian Slope East

Five thermospore OTUs were identified in sediment from Scotian Slope East (Table 4.2). While sulfate is not reduced in any incubations of this sediment (Figure S4.4a), replicate 3 shows evidence of microbial activity by a decrease in formate and lactate concentration over 28 days of incubation (Figure S4.4b). An increase in *Firmicutes* relative abundance occurs in replicates 1 and 3 of 50°C incubations from this site, but only in replicate 3 does *Firmicutes* represent the majority of the relative abundance (Figure S4.4c). The *Firmicutes* community of replicate 3 is composed of OTU 2 (*Symbiobacterium ostreiconchae*), OTU 5 (*Desulfotomaculum tongense*), and OTU 21 (*Aneurinibacillus thermoaerophilus*) while the *Firmicutes* community of replicate 1, which makes up a much smaller proportion of the relative abundance, is composed of

completely different thermospore OTUs, OTU 46 (*Bacillus thermoamylovorans*) and OTU 4497 (*Bacillus hisashii*) (Figure S4.4d).

4.4.1.4 Labrador Shelf

Table 4.2 shows that seven thermospore OTUs were identified in 50°C incubations of Labrador Shelf sediment, one of which, OTU 46 (*Bacillus thermoamylovorans*), was also identified in incubations of Scotian Slope East sediment. Figure S4.6 shows that while sulfate reduction only happens in replicate 3 of Labrador Shelf sediment incubations, decreases of organic acid concentration are observed in each of the replicates. Replicate 1 shows a decrease in lactate and butyrate concentration, and to a lesser extent, a decrease in propionate concentration over the 28-day incubation. Replicate 2 shows a decrease in propionate and butyrate over the incubation period. Replicate 3, the only replicate in which sulfate reduction was observed shows an almost complete decrease in formate, propionate, and butyrate to 0 mM during the 50°C incubation. The decrease in propionate in replicate 1 should not be considered caused by the sediment incubation because the sediment-free, media-only control for this incubation experiment showed a similar decrease in propionate in one replicate over the incubation period (Figure S4.10b). An increase in *Firmicutes* relative abundance is observed in all 3 replicates of Labrador Shelf incubations (Figure S4.6d). The *Firmicutes* community in all three replicates contained OTU 34 (*Caloranaerobacter ferrireduces*) after 50°C incubation. Replicate 1 also contained OTU 89 (*Clostridium halophilium*). Replicates 2 and 3 contained OTU 46 (*Bacillus thermoamylovorans*). OTU 220 (*Defluvitalea saccharophila*) was present in replicate 2 while OTU 469 (*Clostridium caminithermale*) was present in replicate 3 (Table S4.1).

4.4.1.5 Frobisher Bay

Twelve thermospore OTUs were identified in 50°C incubations of sediment from Frobisher Bay (Table 4.2). Replicates 2 and 3 of incubations of sediment from Frobisher

Bay showed sulfate reduction by 7 days of 50°C incubation (Figure S4.7a-b). These 2 replicates also show substantial decreases in organic acid concentration. All 5 organic acids decrease in concentration to, or nearly to, 0 mM after 28 days in replicate 2 while all organic acids except succinate follow the same trend in replicate 3 (Figure S4.7c). Sulfate reduction was not detected in replicate 1 but decreases in formate, lactate, propionate, and butyrate were detected over the 28-day incubation in this replicate. A decrease in propionate was also observed in one replicate of the medium only control incubations (Figure S4.10b), hence observations of propionate decrease warrants caution in replicates 1, 2 and 3 of Frobisher Bay incubations. Increases in *Firmicutes* relative abundance was observed in all three replicates of Frobisher Bay incubations (Figure S4.7d). Each of the replicates had very different *Firmicutes* communities. Replicate 1 was primarily dominated OTU 2 (*Symbiobacterium ostreiconchae*) and secondarily OTU 4 (*Therminicola carboxydiphila*) followed by OTU 55 (*Caldibacillus debilis*). The *Firmicutes* community of replicate 2 consists of OTU 2 (*Symbiobacterium ostreiconchae*), OTU18 (*Desulfotomaculum geothermicum*), OTU 17 (*Bacillus kyonggiensis*), OTU 74 (*Desulfotomaculum reducens*) (Day 14), OTU 1146 (*Bacillus oceanisediminis*), OTU 2250 (*Thermoactinomyces intermedius*), and OTU 55 (*Caldibacillus debilis*). Replicate 3 was dominated by OTU 44 (*Tepidibacillus fermentans*), OTU 642 (*Desulfotomaculum geothermicum*), OTU 216 (*Clostridium thermosuccinogenes*), and OTU 55 (*Caldibacillus debilis*) (Table S4.1). Despite thermospore enrichment in all three replicates there is not a shared OTU present amongst all three. OTU 2 (*Symbiobacterium ostreiconchae*), which was also present in the Scotian Slope East sediment incubations (Table 4.2), and OTU 55 (*Caldibacillus debilis*) are each present in two of the three replicates while all the other OTUs indicated in Table 4.2 are only present in a single replicate.

4.4.1.6 Davis Strait

Sulfate reduction and organic acid concentration decrease were not observed in incubations of sediment from Davis Strait yet a single thermospore, OTU 46 (*Bacillus*

thermoamylovorans), was identified in 50°C incubatons of sediment from this site (Table 4.2). Sulfate concentrations were unstable during the first 9 days of the 50°C incubation but after 30 days the sulfate concentration remained at levels similar to those measured at day 0 (Figure S4.8b). Given that these microcosms only received 0.1 mM of the formate, lactate, acetate, succinate, propionate, and butyrate (a 10-fold reduction in the organic acid concentration compared to the other microcosms, section 4.3.2) only 0.875 mM of sulfate could be reduced assuming that all the organic acids were utilized in sulfate reduction. This provides a possible reason to for the lack of sulfate reduction in incubations of this sediment. Despite the lack of evidence of microbial activity, there was a small increase in *Firmicutes* relative abundance after 14 days of incubation at 50°C (Figure S4.8c). The *Firmicutes* relative abundance was composed almost entirely of OTU 46 (*Bacillus thermoamylovorans*) (Figure S4.8d), an OTU also present in incubated sediment from the Scotian Slope East site and the Labrador Shelf (Table 4.2).

4.4.1.7 Pond Inlet

Sulfate reduction did not occur in 50°C incubations of sediment from Pond Inlet nor was there a notable decrease in organic acids (Table 4.1). A decrease in propionate concentration was observed in replicate 1 of the Pond Inlet sediment incubations but this observation must be treated with caution since a medium only control for this incubation experiment also showed a decrease in propionate during the incubation (Figure S4.10b). *Firmicutes* relative abundance increased slightly after 50°C incubation in replicates 1 and 2 of the Pond Inlet incubations (Figure S4.9c). This increase in *Firmicutes* abundance is primarily due to an increase in OTU 46 (*Bacillus thermoamylovorans*) relative abundance (Figure S4.9d). OTU 46 was also identified in incubated sediment from the Labrador Shelf, Davis Strait and Scotian Slope East (Table 4.2).

4.4.1.8 Baffin Bay North

There was no sulfate reduction or decrease in organic acids in incubations of sediment from Baffin Bay North, nevertheless eight thermospore OTUs were identified (Table 4.1 and 4.2). As observed in in Davis Strait incubations (Figure S4.8), the sulfate concentration in Baffin Bay North sediment incubations were variable over the incubation period but after 30 days of incubation at 50°C, no sulfate had been reduced in any of the replicates (Figure S4.11b). While there is a decrease in sulfate concentration after 30 days compared to the Day 0 sulfate measurement, the large variability of sulfate concentration during the first 9 days of the incubation suggest that this high sulfate measurement at Day 0 may be an artifact of the subsampling or sulfate measurement procedure (Figure S4.11a-b). As discussed in sections 4.3.2 and 4.4.1.6, sediment incubations of Baffin Bay North received 10-fold lower concentrations of the 6 organic acids thus providing lower concentrations of electron donors for sulfate reduction. Organic acids concentration was very low from the onset of the incubation and no detectable organic acid concentration decrease was observed (Figure S4.11c). There was a small increase in *Firmicutes* relative abundance after 14 days of incubation at 50°C even though there was no other evidence of microbial activity (Figure S4.11d). There were 8 thermospore OTUs identified in this incubation, all in <4% relative abundance (Figure 4.11e). OTU 2 (*Symbiobacterium ostreiconchae*) was also identified in sediment from Frobisher Bay and Scotian Slope East. OTU 642 (*Desulfotomaculum geothermicum*) was also found in sediment from Frobisher Bay (Table 4.2). OTU 46 (*Bacillus thermoamylovorans*), which was also identified in sediment from Scotian Slope East, Labrador Slope, Davis Strait, and Pond Inlet was also found in this sediment (Table 4.2).

4.4.1.9 Svalbard

Svalbard sediment incubations (described in Chapter 3) show sulfate reduction and a decrease in lactate, succinate, propionate and butyrate concentration over the 53-day incubation (Figure S4.13a-b). An increase in *Firmicutes* relative abundance is

observed in all 3 replicates of Svalbard sediment incubations after 50°C incubation and 15 thermospore OTUs were identified (Table 4.1 and 4.2). The taxa making up the *Firmicutes* community is very similar between the 3 replicates and contains 15 OTUs. Other than the minor differences in relative abundance of each taxa in the replicates the only other difference between the replicates is that replicate 3 contains OTU 196 (*Halothermothrix orenii*) while replicates 1 and 2 do not contain this OTU in their amplicon libraries (Figure S4.13e). Sediment from Svalbard shares thermospore OTU 136 (*Desulfotomaculum peckii*) with sediment from Baffin Bay North and shares OTU 469 (*Clostridium caminithermale*) with sediment from the Labrador Shelf (Table 4.2).

4.4.2 Shared thermospore OTU identification

Table 4.3: Sediment site type, connected to North Atlantic subpolar gyre (CS) or not connected to North Atlantic subpolar gyre (NCS), and the number of thermospore OTUs identified after incubation at 50°C for 14 days.

Sediment site	Site type	Thermospore OTU count
Scotian Slope West (SSW)	CS	4
Scotian Slope Centre (SSC)	CS	1
Scotian Slope East (SSE)	CS	5
Svalbard (SV)	CS	15
Labrador Shelf (LS)	CS	7
Baffin Bay North (BBN)	NCS	8
Pond Inlet (PI)	NCS	1
Davis Strait (DS)	NCS	1
Frobisher Bay (FB)	NCS	12

Thermospore OTU(s) were identified in 50°C incubations of each of the nine sediment locations in the North Atlantic (Table 4.2) based on the thermospore identification criteria described in Section 2.4.3. Nine of 41 OTUs were identified in

incubations from more than one sediment location; these shared OTUs are listed in Table 4.4 and their phylogenetic relationship is shown in Figure 4.2. Only two shared OTUs were present in more than two sediment locations. OTU 46, which was 97% similar to *Bacillus thermoamylovorans* over a 350 bp region of the V3-V4 region of the 16S rRNA gene based on BLAST search (Madden, 2002) sequence similarity (as described in Section 2.4.3), was found in five of the nine locations – the most of any of the identified OTUs. OTU 2, which was 100% similar to *Symbiobacterium ostreiconchae* over a 350 bp region of the V3-V4 region of the 16S rRNA gene based on BLAST search (Madden, 2002) sequence similarity, was found in three of the nine locations. Seven more thermospore OTUs were found in two locations and 32 were found in only one location.

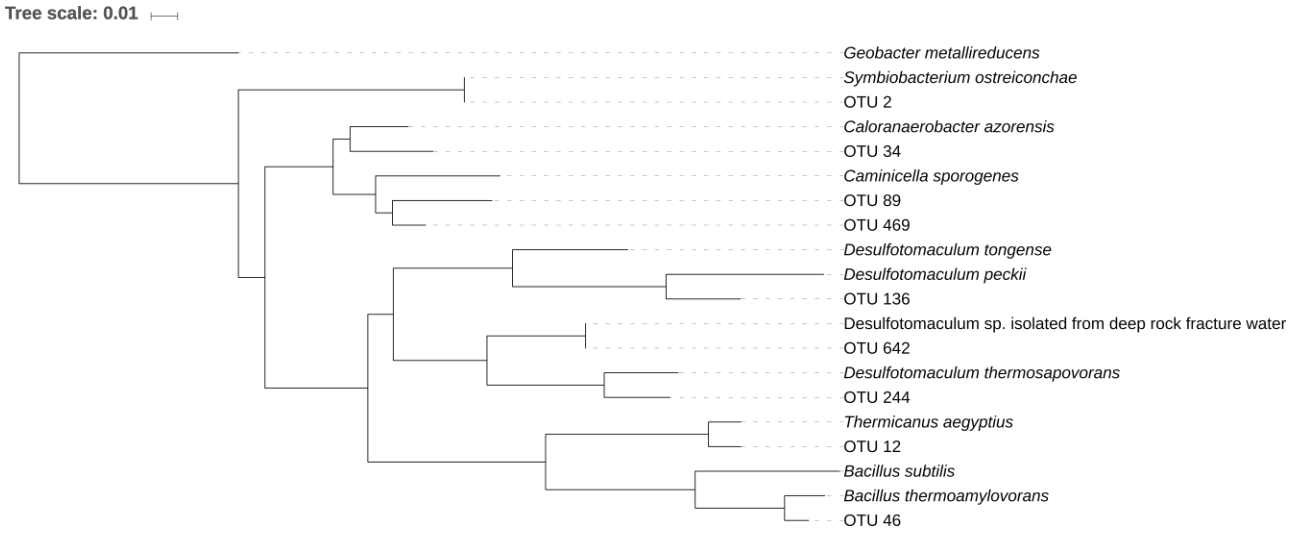

Figure 4.2: Phylogeny of the nine shared thermospore OTUs based on Maximum Likelihood phylogeny estimation. Scale bar identifies 1% sequence divergence. Deltaproteobacterial *Geobacter metallireducens* is included as an outgroup.

Table 4.4: Thermospore OTU identification based on BLAST 16S rRNA gene sequence similarity. Sites from which each thermospore OTU was identified after 50°C incubation are listed. Of forty-one thermospore OTUs identified, only the nine that are identified in sediment from more than one location are listed.

Shared OTU	BLAST ID	% ID	Accession number	Sites
OTU 2	*Symbiobacterium ostreiconchae*	100	NR_134208.1	SSE FB BBN
OTU 12	*Thermicanus aegyptius*	100	NR_025355.1	SSC SSW
OTU 34	*Caloranaerobacter ferrireducens*	97	NR_135860.1	LS SV
OTU 46	*Bacillus thermoamylovorans*	97	NR_117028.1	SSE BBN PI DS LS
OTU 89	*Clostridium halophilum*	96	NR_125713.1	LS SV
OTU 136	*Desulfotomaculum peckii*	94	NR_109724.1	BBN SV
OTU 244	*Desulfotomaculum thermosapovorans*	96	NR_119247.1	SSW SV
OTU 469	*Clostridium caminithermale*	97	NR_041887.1	LS SV
OTU 642	*Desulfotomaculum geothermicum*	95	NR_042044.1	FB BBN

4.4.3 Thermospore OTU occurrence and site connection to North Atlantic subpolar gyre

The average number of thermospore OTUs identified in sites connected to the North Atlantic subpolar gyre (CS) and those not connected to the North Atlantic subpolar gyre (NCS) were not notably different with averages of 6.4 and 5.5 respectively (Figure 4.4). This study suggests that a sediment site's connectedness to the North Atlantic subpolar gyre does not increase the number of thermospores in the sediment, although the number of sites (nine) and the number of thermospore OTUs identified (forty-one) is likely too low to make this study conclusive (chi-squared statistics were unable to be done on the data owing to the high number of samples with less than five OTUs)

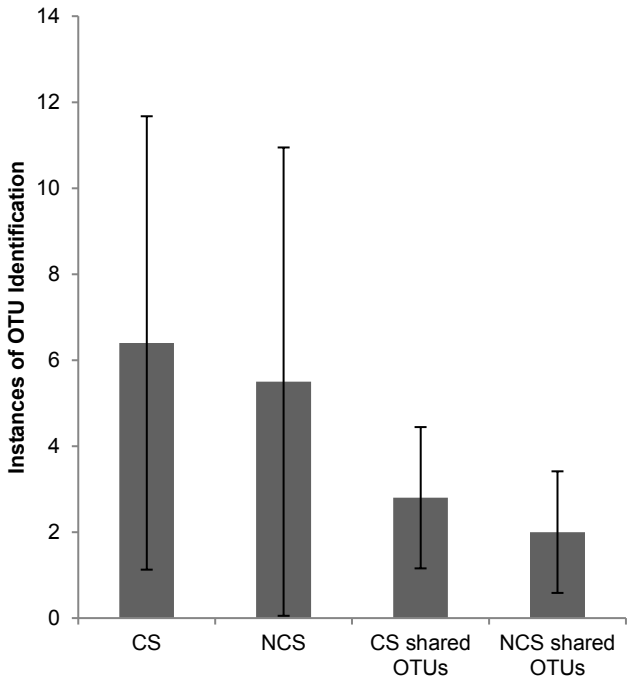

Figure 4.3: Average number of instances of thermospore OTU identification or instances of shared OTU identification at sediment sites connected to the North Atlantic subpolar gyre (CS) and sediment sites not connected to the North Atlantic subpolar gyre (NCS). An "incidence of thermospore OTU identification" refers to the number of thermospore OTU that are identified at a particular location. An "incidence of shared OTU" refers to the number of OTUs at a particular location that are also found at one or more other locations. The same OTU may be found in more than one sediment location and is counted as an incidence of OTU identification at each location. "CS shared OTUs" refers to the number of instances of shared OTUs counted at all CS sites. "NCS shared OTUs" refers to the number of instances of shared OTUs counted at all NCS sites. Error bars show standard error in the number of OTUs identified among the sites or the number of shared OTU instances among the sites. CS represents "Connected Sites" and refers to the sites in the North Atlantic that are connected to the North Atlantic subpolar gyre or the Norwegian Current. NCS represents "Not Connected Sites" and refers to the sites in the North Atlantic that are not connected to the North Atlantic subpolar gyre or the Norwegian Current.

Thermospores have been found in the water column above sediments from which they are enriched in incubation experiments (de Rezende *et al*., 2013; Volpi *et al*., 2017) therefore, a sediment site may be expected to share a thermospore OTU with another sediment site if there are water currents connecting the two sites. Most of the thermospore OTUs identified in this study are found in a single sediment location (Figure 4.5). While there are nine OTUs that are shared between different locations (indicated in Figure 4.5 as OTUs "shared between one or more sites") the number of enriched OTUs found at one site that are also found in another location (shared OTUs) is not significantly different between sites connected to the North Atlantic subpolar gyre (CS), and those not connected to the North Atlantic subpolar gyre (NCS) with 2.8 and 2.0 incidences of shared OTUs identified respectively (Figure 4.4). (Note that an "instance of a shared OTU" differs slightly from simply a "shared OTU"; while "shared OTU" refers to the biogeography of the OTU that is detected – one that is identified at more than one site – "incidences of a shared OTU" refers to the event of a shared OTU being detected at a site, thus a single OTU that is found in two locations will be counted as two incidences of shared OTU identification. For example, in this study, there are 22 instances of a shared OTU being detected but there are only 9 shared OTUs) These data suggest that a site's connectedness to the North Atlantic subpolar gyre does not increase the number of incidences thermospore OTUs detected in the sediment will also be found in another location. A site connected to the North Atlantic subpolar gyre (CS) appears no be more likely to hold an OTU that is also found in another location than a site that is not connected to the North Atlantic subpolar gyre (NCS).

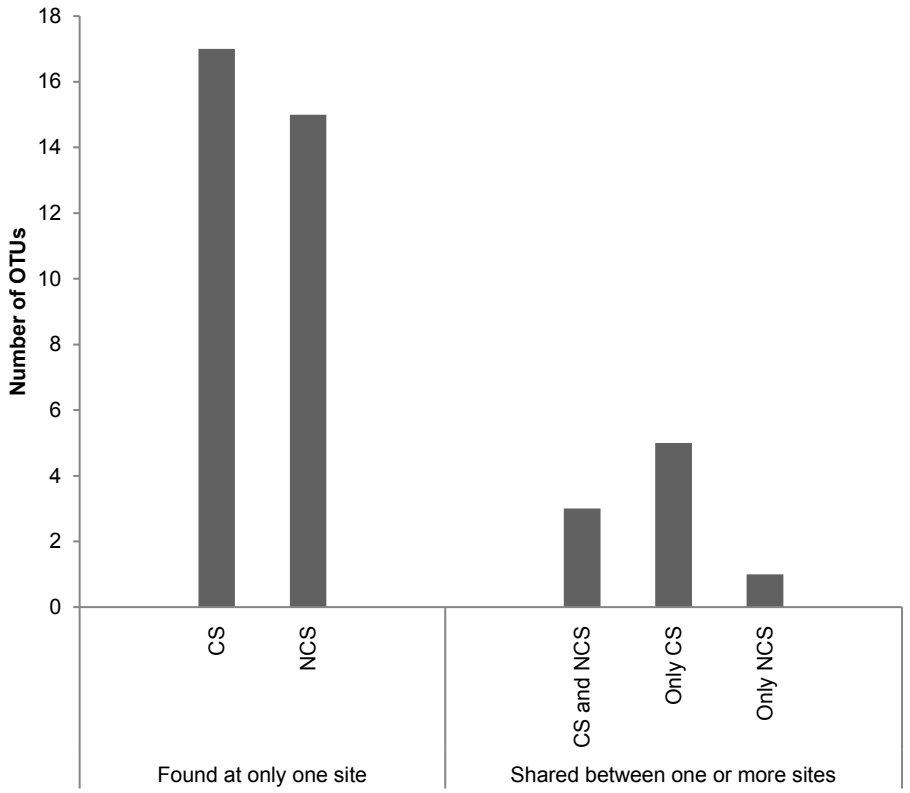

Figure 4.4: The identified thermospore OTUs are classified into 2 categories: those found at only one location and those that are present in more than one location (shared). OTUs are further categorized based on their site's connectedness to North Atlantic subpolar gyre: OTUs only found at one CS site or one NCS site, OTUs that are shared and found at CS and NCS sites, OTUs that are shared and found at CS sites only, or OTUs that are shared and found at NCS sites only. CS represents "Connected Sites" and refers to the sites in the North Atlantic that are connected to the North Atlantic subpolar gyre or the Norwegian Current. NCS represents "Not Connected Sites" and refers to the site in the North Atlantic that are not connected to the North Atlantic subpolar gyre or the Norwegian Current. All 41 thermospore OTUs detected in this study are represented in this chart.

4.4.4 Assessing thermospore OTU absence in the largest 16S rRNA gene library

Since the detection and identification of thermospore OTUs is dependent on their germination and growth, the lack of a particular OTU's detection does not confirm its absence *in situ* at the sediment location. The largest amplicon library in this study (2 157 989 reads passing quality control) came from Frobisher Bay incubation replicate 1 after 14 days of incubation (Figure 2.1). Only two thermospore OTUs were identified from this sample according to the cut off criteria discussed in section 2.4.3. However, this library also contains reads in abundance between 3 and 138 of other thermospore OTUs identified in other locations. If the presence of these reads, despite being in number far below the 0.16% error threshold described by Nelson *et al*. (2014), could allow for these taxa to be considered identified thermospore OTUs in this sample, then only five more OTUs would be added to the number of shared OTUs, bringing the total number of shared OTUs in the dataset to fourteen. Of the forty-one thermospore OTUs detected, twenty-seven are not found in the largest library in even a single read.

4.5 Discussion

Incubations (50°C) of sediment from nine different locations in the North Atlantic led to the enrichment and identification of forty-one thermospore OTUs, nine of which were detected in more than one sediment location. A thermospore phylotype was present in at least one biological replicate from each sediment location implying that thermospore presence is common to North Atlantic sediments despite temperatures that are too low to allow for thermospore germination. Since temperatures are too low for growth of thermophilic endospore-forming bacteria, their presence in the sediment as dormant endospores is presumably owing to passive dispersal from warm locations (Hubert *et al*., 2009; Hubert and Judd, 2010). This passive dispersal would rely on water movements of which ocean currents may be a primary driver.

Sediment sites connected to the North Atlantic subpolar gyre did not have a significantly higher number of thermospore OTUs nor did these sites have a significantly higher detectable number of shared thermospore OTUs. Likewise, thermospore OTUs present in 50°C incubations from sites connected to the North Atlantic subpolar gyre were not more likely to be found in other connected sites (CS) than unconnected sites (NCS). This study suggests that connectedness to the subpolar gyre, and the Mid-Atlantic Ridge hydrothermal systems that it traverses, does not affect the dormant thermospore community in a way detectable by the methods employed in this study. A larger number of sediment sites and incubation conditions that promote the germination of a more diverse population of thermophilic endospore-formers, eg. the addition of aerobic incubations (see Chapter 3) and/or media more selective for fermentative physiology (Volpi *et al*., 2017) that may lead to a larger number of identified thermospore phylotypes per site may provide data that could make these findings conclusive.

The results of this study showing no notable difference in thermospore numbers (shared and not-shared OTUs) may suggest that thermospores deposited into marine sediment may be coming from sources other than and/or in addition to the Mid-Atlantic Ridge spreading centers. Alternatively, the dispersal of thermospores from the Mid-Atlantic Ridge spreading centers may extend far beyond those sites connected to the North Atlantic subpolar gyre and the other currents immediately connected to the North

Atlantic Current (Norwegian Current) to more distant sediments. Warm environments other than those of the of the hydrothermal vent systems at the Mid-Atlantic Ridge include oil production facilities (Magot, 2005) such as those found in the Barents Sea or spreading centers and hydrothermal vent systems such as those found along the Gakkel ridge in the Arctic Ocean (Edmonds *et al*., 2003). Similarly, cold seeps, which may be connected to deeper and warmer hydrocarbon reservoirs, are a possible source of thermophiles (Hubert and Judd, 2010). These warm sites are also connected to the water column above the sediment by seeping fluids and could similarly be a source of thermospores. Like the North Atlantic Current and its connection to the Mid-Atlantic Ridge system, the Arctic Ocean has currents connecting the Gakkel ridge to other sites within the Arctic Ocean as well as to the northern currents of the North Atlantic (Nøst and Isachsen, 2003); these warm locations are other possible source habitats of thermospores found in sediment of the North Atlantic.

Thermospore OTUs of particular interest in this study were those that were present in more than one sediment location. Of these, two OTUs were present in more than two locations – OTU 46 was present in five locations and OTU 2 was present in three locations. Interestingly, OTU 46 shared a phylogenetic lineage with *Bacillus thermoamylovorans*, an endospore forming *Bacilli*. Endospores, specifically those of *Bacillus subtilis*, have long been used to study the possibility of panspermia, the theory that life is dispersed throughout the universe (Weber and Greenberg, 1985; Horneck, 1993; Nicholson *et al*., 2000; Nicholson, 2009). Radiation is a primary stress to life exposed to space conditions and spores are known to have an increased tolerance to radiation and other stressors, eg. temperature extremes, pressure extremes, and tolerance to toxins (Setlow, 1994). In addition, very slow decay rates allow endospores to remain viable for thousands to millions of years (Cano and Borucki, 1995; Vreeland *et al*., 2000; de Rezende *et al*., 2013) and it has been proposed that thermophilic endospores may remain viable for even longer (Nicholson, 2003). The tolerance to radiation and other stressors and viability after very long periods of dormancy make endospore-formers prime candidates to studying the feasibility of interplanetary/interstellar microbial dispersal. The phylogenetic proximity of OTU 46 to species known to survive the conditions of space highlights the importance of

understanding the dispersal capabilities of these bacteria to prevent the accidental occurrence of panspermia during human space exploration.

OTU 2 shows close phylogenetic relationship to *Symbiobacterium ostreiconchae* of the *Symbiobacterium* lineage of the phylum *Firmicutes*, a syntrophic organism which requires commensalism with *Geobacillus stearothermophilus* or *Ureibacillus* spp. (Ohno *et al*., 2000, Sugihara *et al*., 2008; Shiratori-Takano *et al*., 2014). *S. ostraconchae*, *G. stearothermophilus*, and *Ureibacillus* sp. are moderately thermophilic and produce endospores (Fortina *et al*., 2001; Nazina *et al*., 2001; Shiratori-Takano *et al*., 2014). Interestingly, there were no occurrences in this study of a thermospore phylotype or a member of the family *Bacillaceae* that was common to all 3 locations in which OTU 2 was found. The syntrophic partner taxon with OTU 2 at its original warm source was either not equally dispersed about the North Atlantic with OTU 2 or OTU 2 is common to multiple warm temperature sources, and living commensally with different *Bacillaceae* sp. Alternatively, the taxon living syntrophically with OTU 2 may lack the endospore phenotype and would thus fail to be detected after the incubation conditions and subsequent implementation of the thermospore identification protocol used in this study.

It is important to note that the OTUs used in this study are grouped based on sharing 97% similarity along the 16S rRNA gene amplicon. Because of this, OTU 2 and OTU 46 may be groupings of many different taxa that happen to share 97% identity along the 16S rRNA gene amplicon. Oligotyping, a technique that allows taxonomy within OTUs to be resolved (Eren *et al*., 2013), would provide a more robust analysis of the presence of shared OTU at different sediment locations and should be done in future biogeographic studies using thermospores.

The absence of a thermospore OTU is difficult to prove since not detecting it based on the criteria in section 2.4.3 simply implies that the organism did not increase in abundance over the course of the 50°C incubation – no claim can be made about its absence *in situ*. Based on the identification method used in this study (section 2.4.3), the best observation of an OTU being absent is if it was missing in the largest 16S rRNA gene amplicon library. Replicate 1 of the 14 day Frobisher Bay incubation sample had over two million reads yet many of the OTUs detected by the criteria in section 2.4.3 in other samples are not present in even a single read in this library.

The presence of thermospore OTUs in some sediments but not in others suggests that these thermospores are not cosmopolitan to the sediment of the North Atlantic, although this cannot be claimed conclusively due to failure to detect spores that did not germinate. Müller *et al*. (2014) demonstrated that thermospore phylotypes are not equally distributed in the global ocean providing some evidence pointing to the endemism of some thermospores; they suggested connectivity to ocean circulation as the cause of differential thermospore presence between sediments, a theory the present biogeographical study cannot corroborate.

Understanding the geophysical processes that contribute to thermospore passive dispersal leads to understanding microbial dispersal beyond thermospores specifically and the unique physiologies associated with thermophily and endospore formation make thermospores seemingly good model organisms to study these processes. Further research on thermospore dispersal may help resolve questions of microbial endemism and further understanding of the factors contributing to the dormant microbial seed bank. Additionally, better knowledge of the dispersal capabilities of endospore-forming bacteria is important to prevent the contamination of potentially sterile sites, a sensational example of this being the prevention of contamination of extraterrestrial environments during human space exploration. However, inconsistencies in OTU presence that arise among replicates cast some doubt on the effectiveness of using germination experiments to trace these dormant members of the rare biosphere and more consistent mechanisms of detection may be required before thermospores can unequivocally be considered model organisms for microbial passive dispersal.